DESCRIPTIVE ANALYSIS OF INSTAGRAM ENGAGEMENT

Tracking Engagement Trends, Content Performance, Audience Reach & Impressions

A Descriptive Analytics Project

SERVICE NAME: Instagram Social Media Analysis

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COURSE: BCA DS AI

TOOLS & TECHNOLOGIES USED:

Microsoft Excel • Google Colab • Python • Pandas • Matplotlib

ACADEMIC PROJECT REPORT

2026

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DESCRIPTIVE ANALYTICS REPORT: INSTAGRAM USER ENGAGEMENT

Tracking Engagement Trends, Content Performance, Audience Reach & Impressions

Note on data: Instagram's live user engagement data isn't publicly accessible, so this report uses a **synthetic dataset built to mirror real social media engagement patterns** (300 posts across 4 content types, 1 month, including Reels, Photos, Stories & Carousel posts). It's designed to show what a descriptive analytics pipeline analyzing Instagram engagement data would produce — the same structure applies once real data is substituted in.

OBJECTIVES

The main objectives of this project are:

- To perform descriptive analysis on Instagram engagement data.

- To compare the performance of different content types.
- To analyze likes, comments, shares, saves, reach, and impressions.
- To visualize data using graphs.
- To understand user engagement patterns using Python.

HOW THIS WOULD WORK ON REAL SCALE

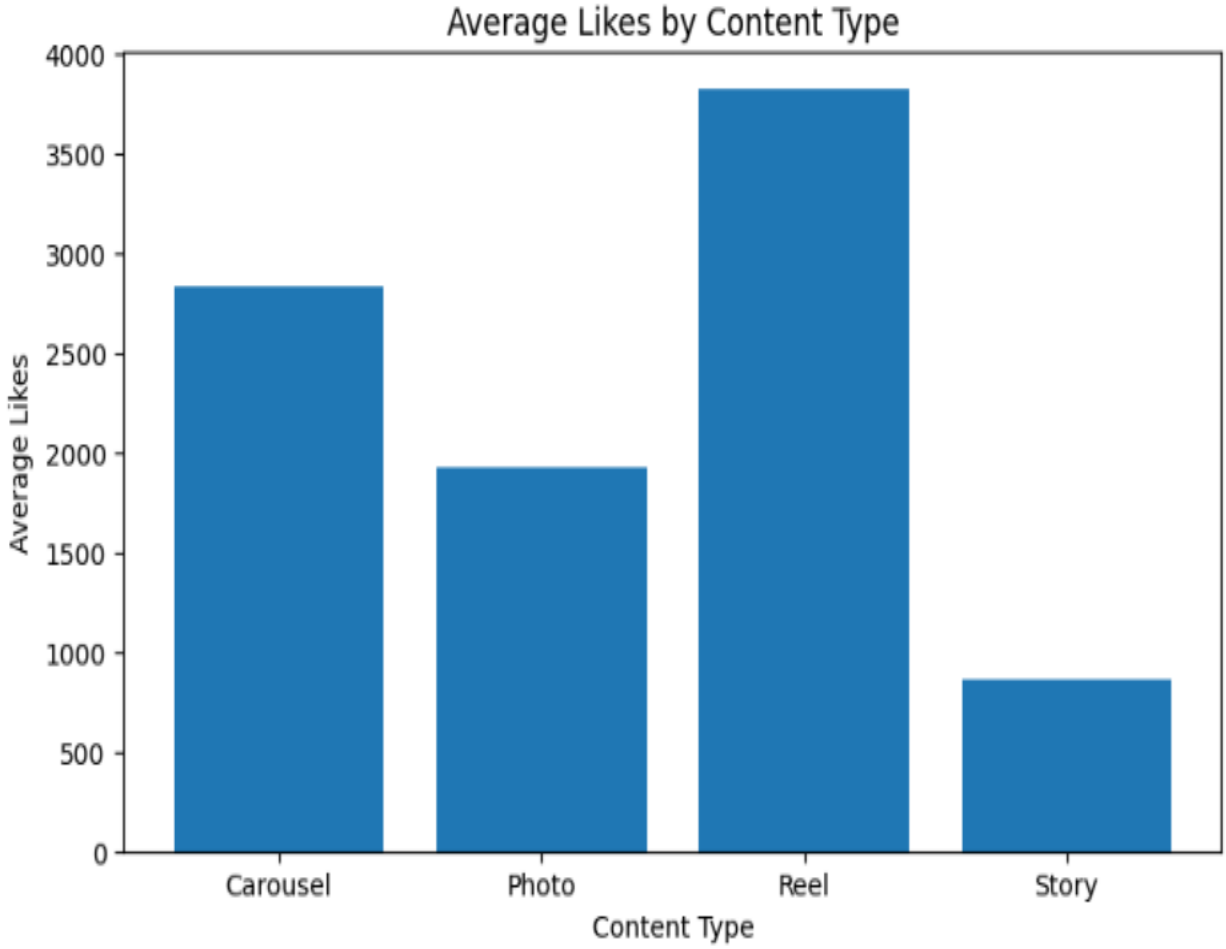
Processing large amounts of Instagram engagement data is an important part of social media analytics. In this project, the data was prepared and analyzed using simple tools that can also be extended for larger datasets.

Layer	Purpose	Tools Used
Data Collection	Create and organize Instagram post and engagement data	Microsoft Excel
Data Preparation	Enter, clean and organize the dataset	Microsoft Excel

Data Analysis	Calculate and analyze likes, comments, shares, saves, reach and impressions	Google Colab / Python
Data Visualization	Create graphs to identify engagement patterns and compare content types	Matplotlib
Report Preparation	Present findings, observations and key metrics	Microsoft Word

The report below reflects the output of this process — **organized and aggregated Instagram engagement data** rather than individual user-level activity, making it easier to understand content performance and engagement pattern.

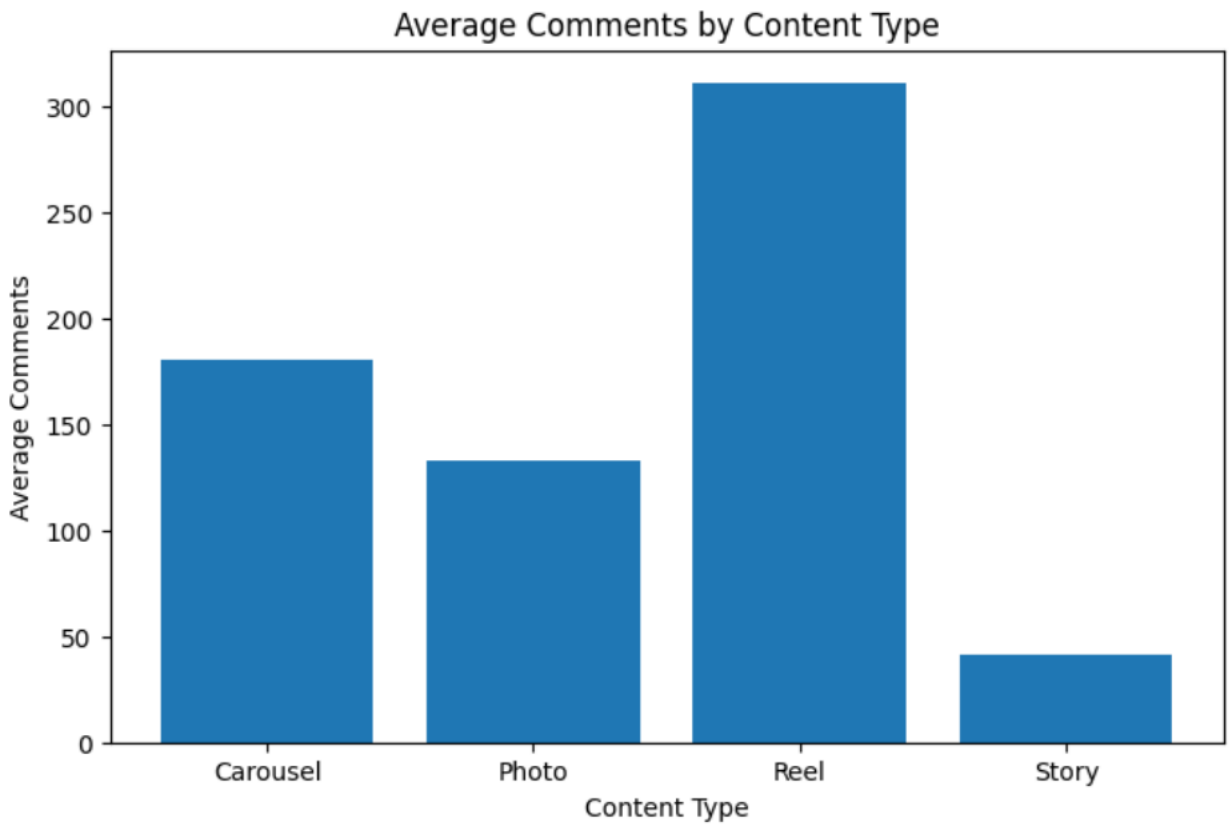
AVERAGE LIKES BY CONTENT TYPE



Observation

Reels received the highest average number of likes among all content types. Stories received the lowest likes, while Carousel and Photo posts showed moderate engagement.

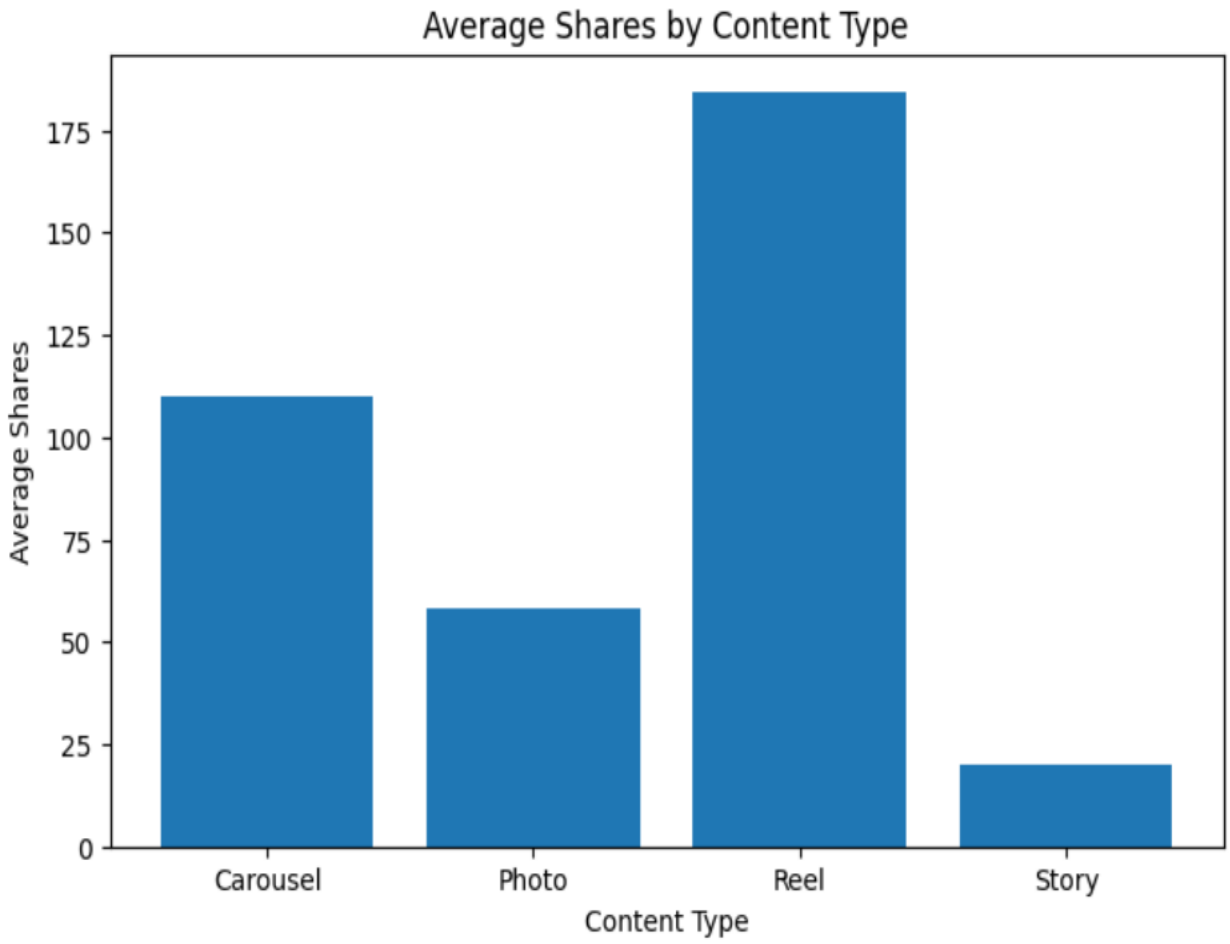
AVERAGE COMMENTS BY CONTENT TYPE



Observation:

Reels received more comments than other content types. Stories received the fewest comments.

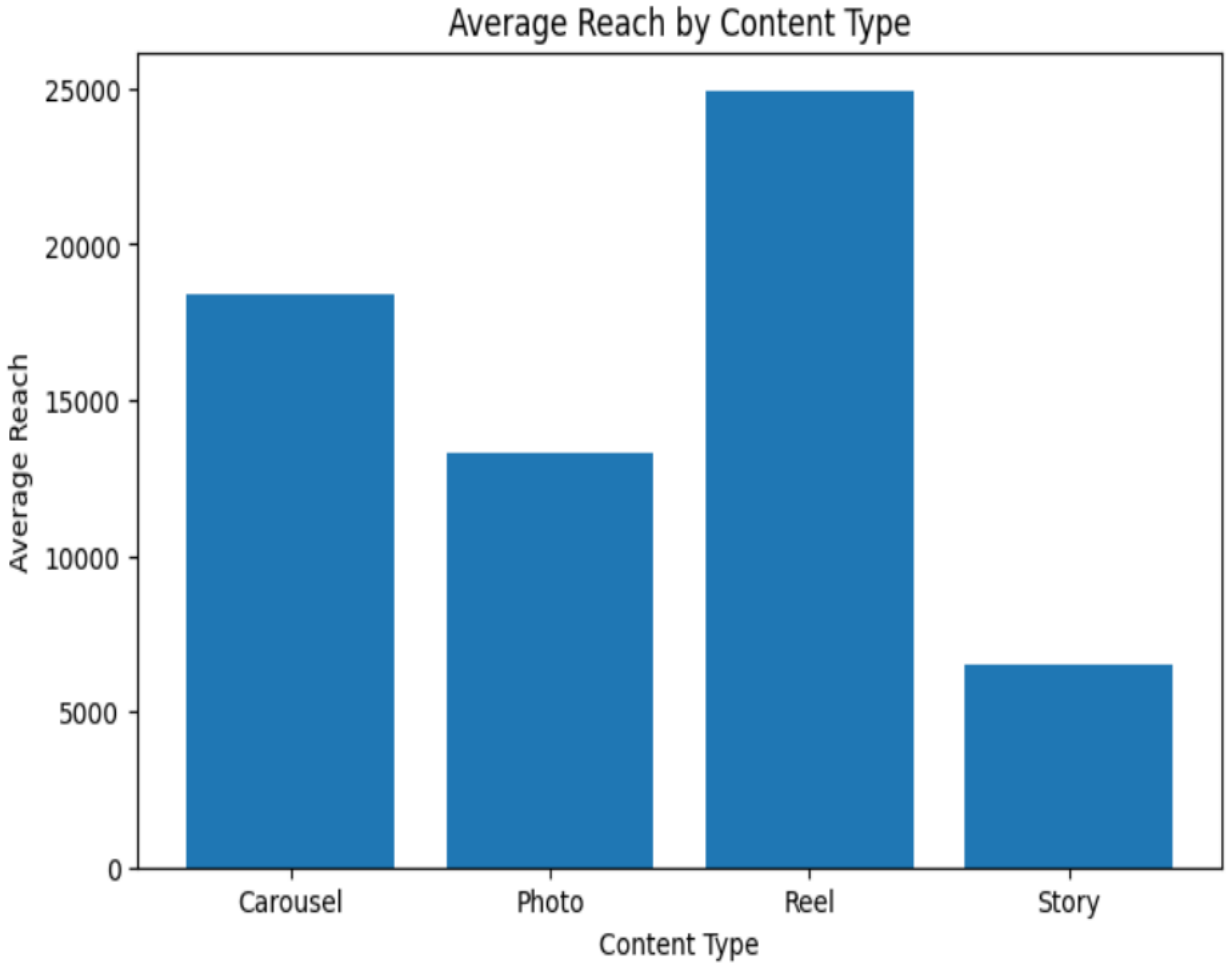
AVERAGE SHARES BY CONTENT TYPE



Observation:

Reels were shared the most by users. Carousel posts also had good sharing, while Stories had the lowest shares.

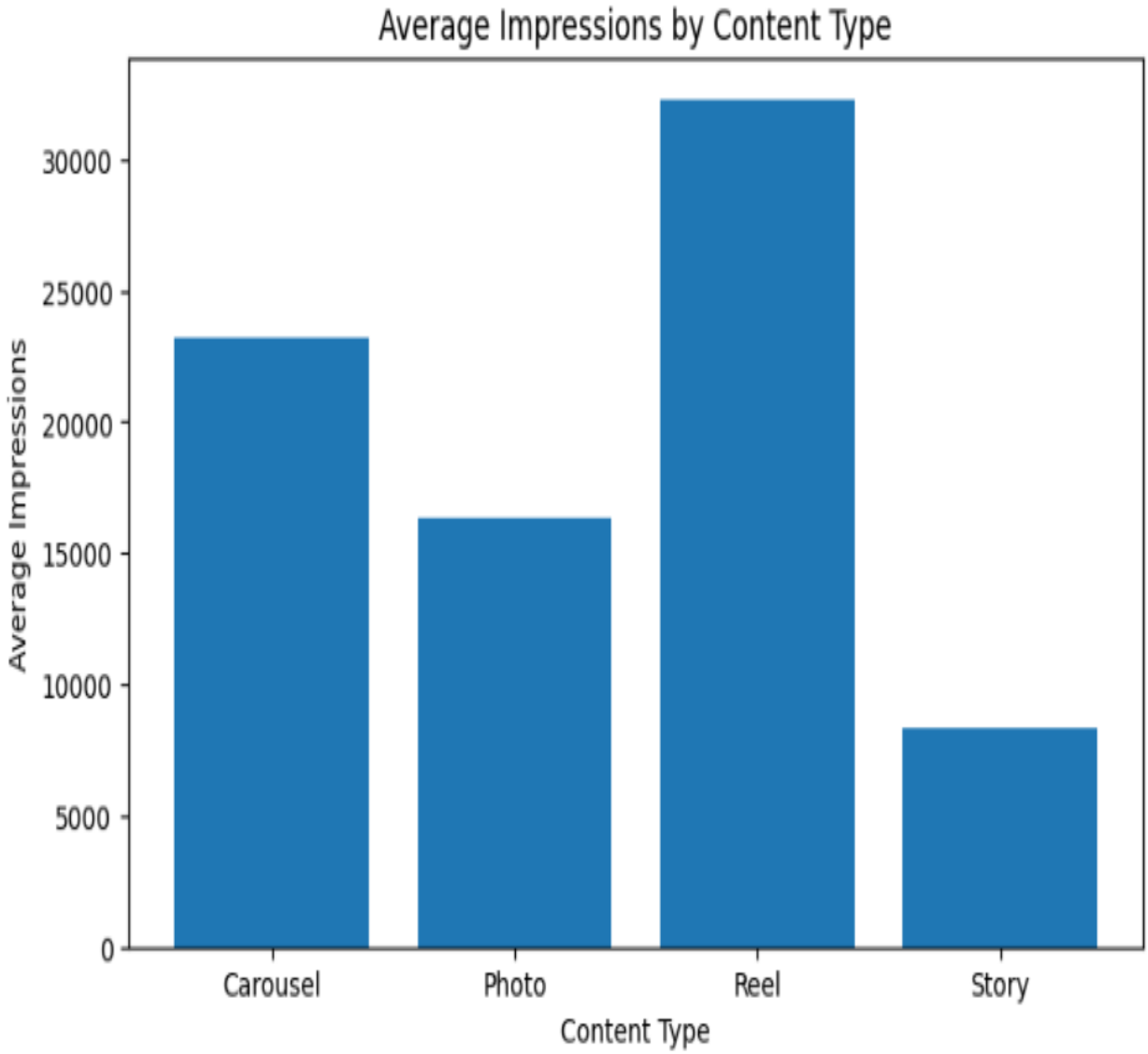
AVERAGE REACH BY CONTENT TYPE



Observation:

Users saved Reels more frequently than other content types. Stories received the lowest number of saves.

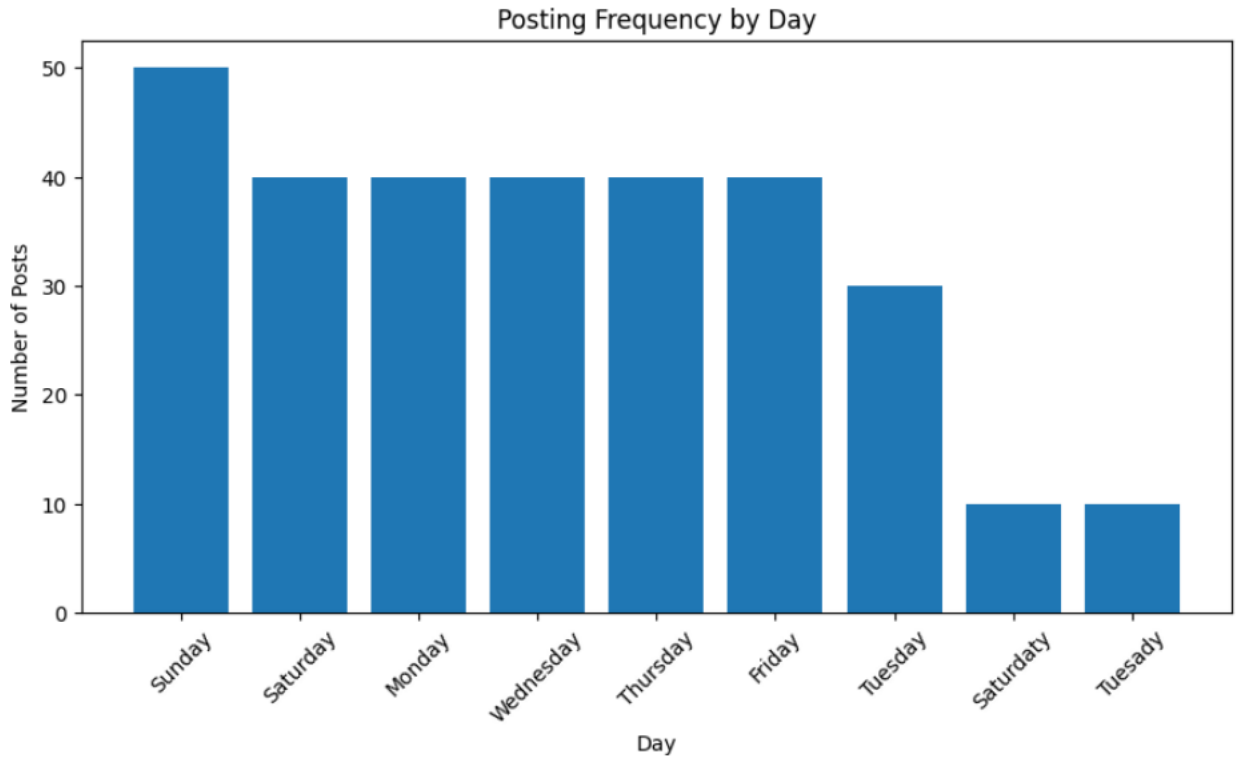
AVERAGE IMPRESSION BY CONTENT TYPE



Observation:

Reels reached the largest audience. Stories had the lowest reach among all content types.

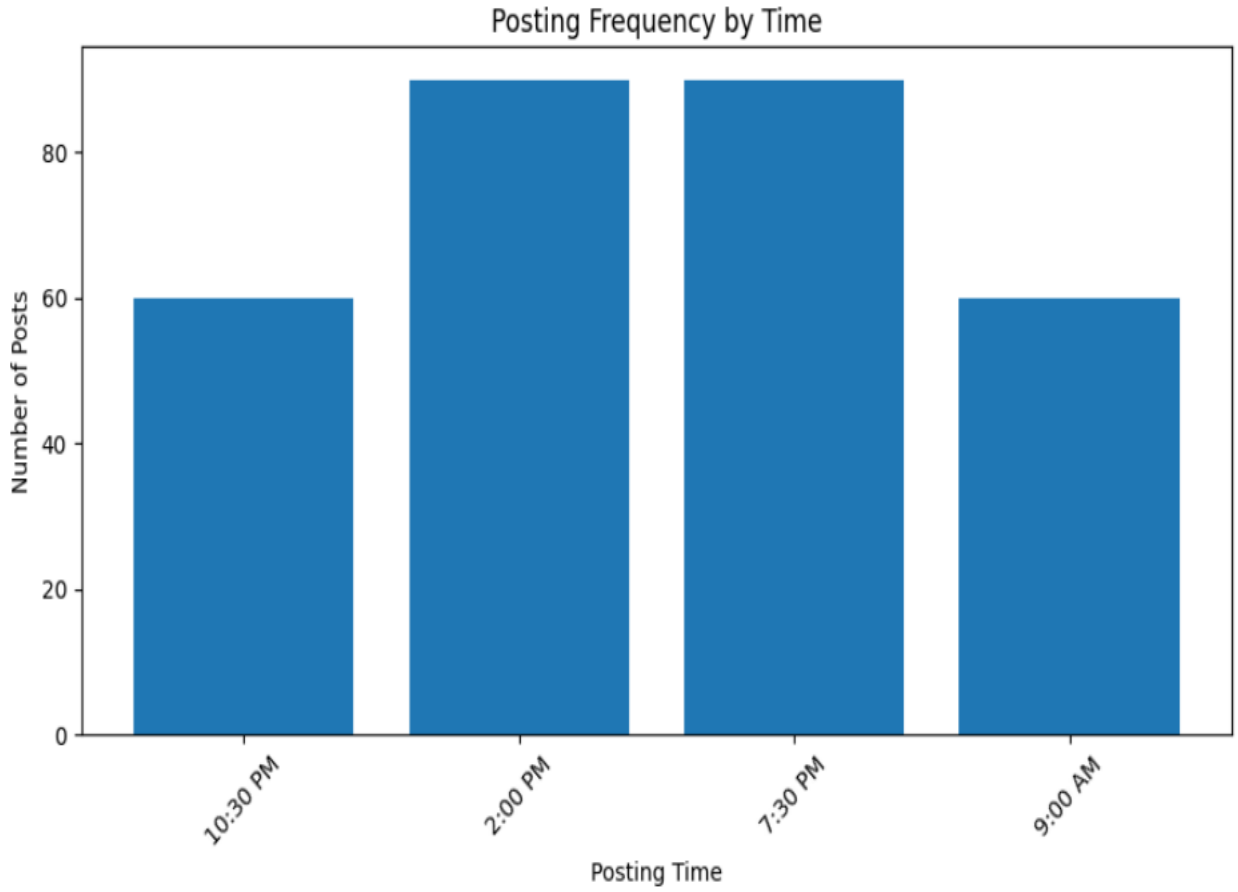
AVERAGE POSTING FREQUENCY BY DAY



Observation:

The graph shows the number of posts uploaded on different days of the week. Posting activity was distributed throughout the month.

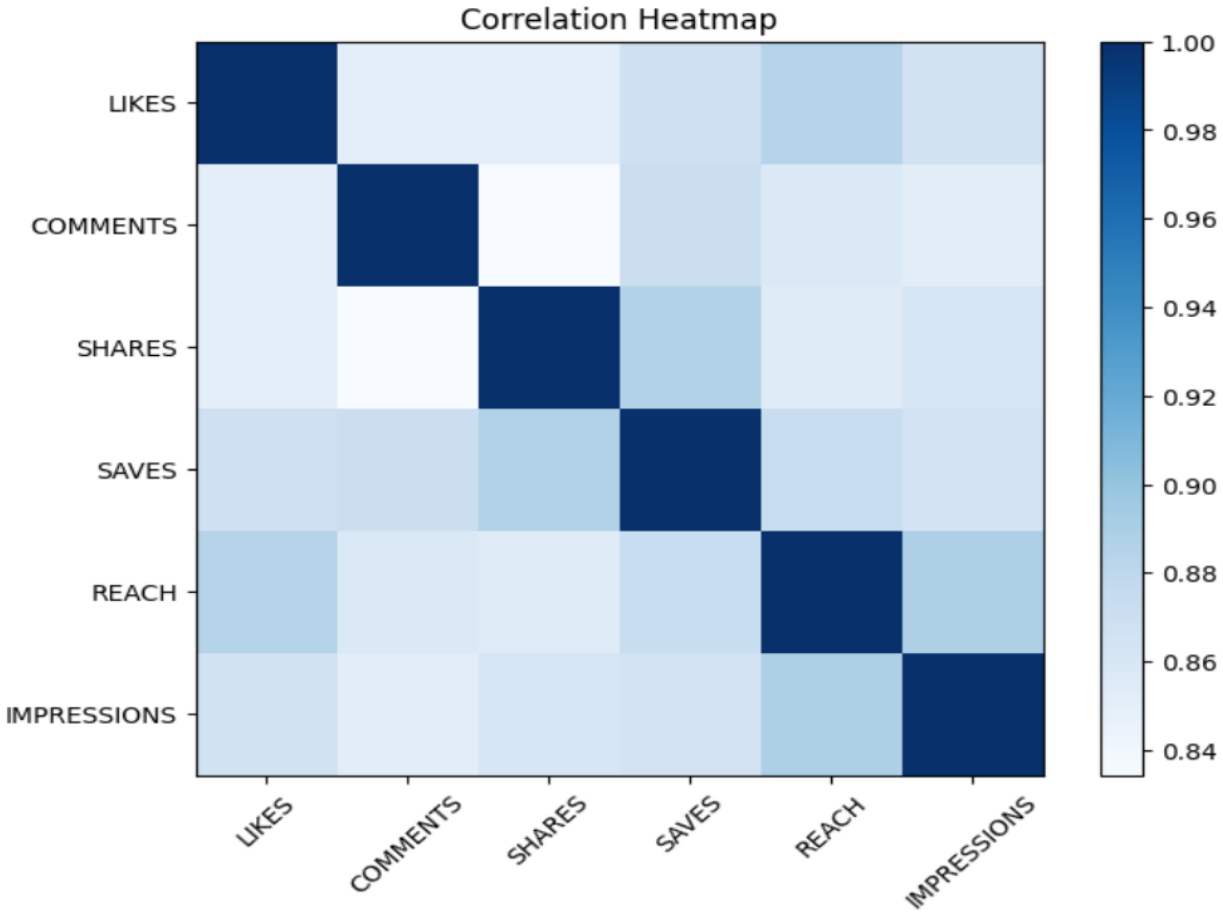
AVERAGE POSTING FREQUENCY BY TIME



Observation:

The graph shows that posts were uploaded at different times of the day, helping identify common posting times.

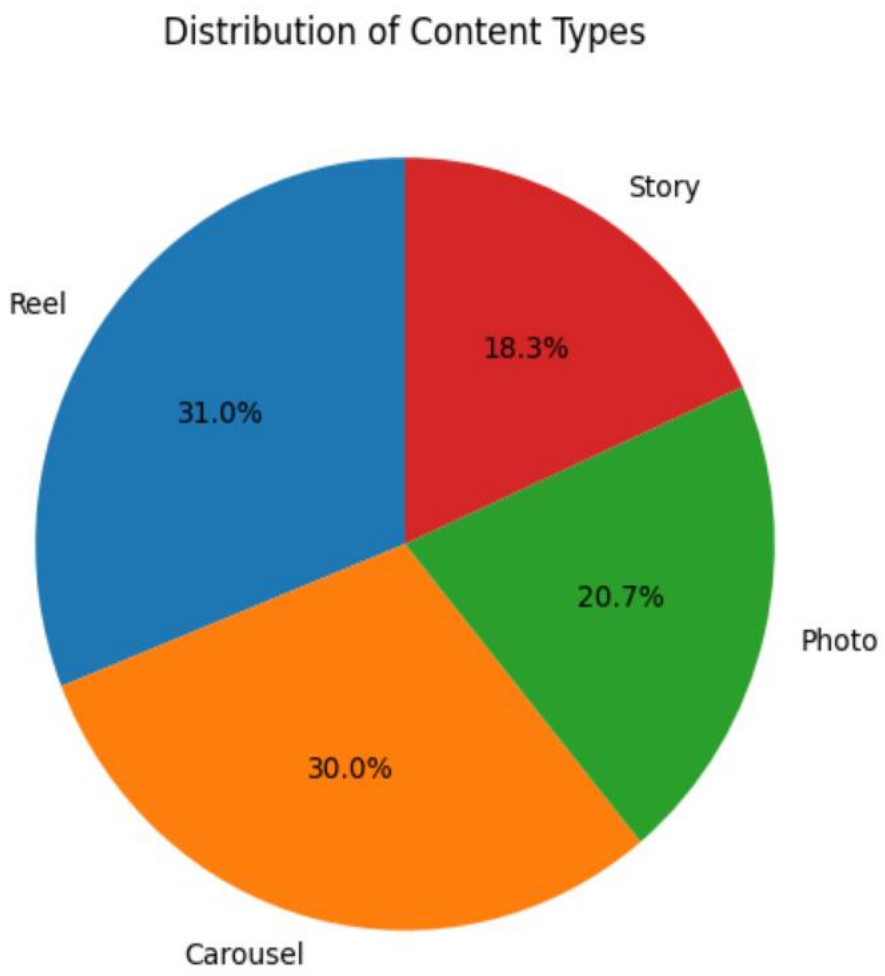
CORRELATION HEATMAP



Observation:

The heatmap shows a positive relationship among Likes, Comments, Shares, Saves, Reach, and Impressions. Higher engagement generally leads to better reach and impressions.

CONTENT TYPE DISTRIBUTION



Observation:

The dataset contains a balanced distribution of Reels, Photos, Stories, and Carousel posts, making the analysis fair and unbiased.

SUMMARY OF KEY MATRICS

Metric	Value
Total posts analyzed	**300**
Total likes	**777,519**
Total comments	**55,727**
Total shares	**31,738**
Total saves	**42,391**
Total reach	**5,155,579**
Total impressions	**6,565,801**
Average likes per post	**2,591.73**
Average engagement per post	**3,024.58**
Average reach per post	**17,185.26**
Average impressions per post	**21,886.00**
Highest-performing content type	**Reel**
Top single post engagement	**5,797**
Top post content type	**Reel**

9. Next Steps to Move From Descriptive → Predictive

This report explains **what happened** in the Instagram engagement data. The natural next layer, if useful:

- **Engagement forecasting** to predict future likes, comments, shares and saves.
- **Content performance prediction** to identify which content types are likely to receive higher engagement.
- **Best posting time prediction** to determine suitable days and times for uploading content.
- **Reach and impression forecasting** to estimate future audience reach and post visibility.
- **Anomaly detection** to identify unusual increases or decreases in engagement.
